

Velocity measurements of bubbles in liquid nitrogen

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Abstract

Cavitation is one of the troublesome problems in rocket turbo pumps, and most of high-efficiency rocket propellants are cryogenic fluids. There is a need for experimental characterization of bubble flows in such conditions. We propose in this study to evaluate the performance of Laser Doppler Velocimetry (LDV) to measure the rising velocity of bubbles in liquid nitrogen N_2 . LDV is compared with two imaging techniques: Particle Tracking Velocimetry (PTV) and high-speed imaging. A cryostat with optical access has been designed to evaluate the feasibility of these optical diagnostics in a cryogenic medium. Comparisons with the three optical diagnostics show a good agreement. Bubble shape characterization, obtained with imaging, represents valuable information to explain the oscillations of velocity. LDV in backward scattering could be used to characterize cavitation in rocket turbo pumps for which optical access is limited.

Introduction

Recent developments of rocket engines require extremely high combustion chamber pressure, resulting in high levels of rotating speed for the turbo pumps. In this case, cavitation can be a serious problem as it induces unstable pressure fluctuations. Furthermore, cavitation bubbles can decrease the efficiency of the pump or even destroy the rotor blades. In rocket turbo pumps, the cryogenic fluid can be liquid oxygen at 90K or liquid Hydrogen 20.4K. The temperature in cavitation region decreases slightly due to latent heat absorption, so the cavitation becomes improved in a cryogenic fluid [1].

There is a need for design criteria of rocket turbo pumps and there are few results in the literature on experiments or calculations on the behavior of bubbles in a cryogenic medium. There is a need for velocity measurements in cryogenic fluids. We propose in this paper to perform velocity measurements on bubbles of nitrogen gas in liquid nitrogen at 77K. The performance of the optical diagnostics is also addressed in this work.

Nagashima et al. conducted both calculations and experiments for liquid nitrogen cavitation in a 2D laval nozzle [1] and around a foil [2]. Those two studies observed numerically and experimentally a temperature decrease and a depression in the region where the cavitation occurs. Cavitation bubbles were also visualized but their velocity was not measured. Ishii et al.[3], [4] performed velocity measurements with a Particle Image Velocimetry system (PIV) to obtain the void fraction in the throat of a venturi channel. They used cavitation bubbles in liquid Helium as tracers for the PIV system.

The problem is that in cryogenic fluids, it is hard to distinguish the gas phase from the liquid phase. The purpose of this work is to study the influence of the cryogenic medium on the performance of a Laser Doppler Velocimetry (LDV). A high-speed imaging system and a Particle Tracking Velocimetry (PTV) device have been also implemented to visualize bubbles in liquid nitrogen in a specially designed cryostat with multiple optical accesses. Imaging systems have been chosen because the temperature of the medium does not influence those techniques. First we propose a simple model to describe the rising velocity of a single bubble in liquid nitrogen. We describe the optical setup of the diagnostics. Then we compare results of the three optical diagnostics and differences serves as a basis to put in focus the limitations of the devices.

Modelling injection of bubbles

The cryostat is divided in two different tanks containing liquid nitrogen at $T_b=77K$ (See Figure 1). The upper tank of volume 2L is independent from the test section and maintains the cryostat at a constant temperature. The lower tank of 0.5L that constitutes the test cell, is filled with liquid N_2 at T_b and P_s , and is equipped with three optical accesses in two perpendicular directions. Quartz windows are used to absorb infrared radiation

coming from the room temperature and to remain transparent to visible light [5]. The whole cryostat is made with a high vacuum shield (approximately 10^{-4} Pa) to reduce thermal loss.

A gas injection device has been designed to seed the test cell with bubbles. N_2 gas is injected through a needle of internal diameter $D_i = 200\mu m$ and gas flow is controlled with a flow meter. Therefore two populations of bubbles are present simultaneously in the test section: bubbles injected through the needle and boiling bubbles of the saturated liquid N_2 .

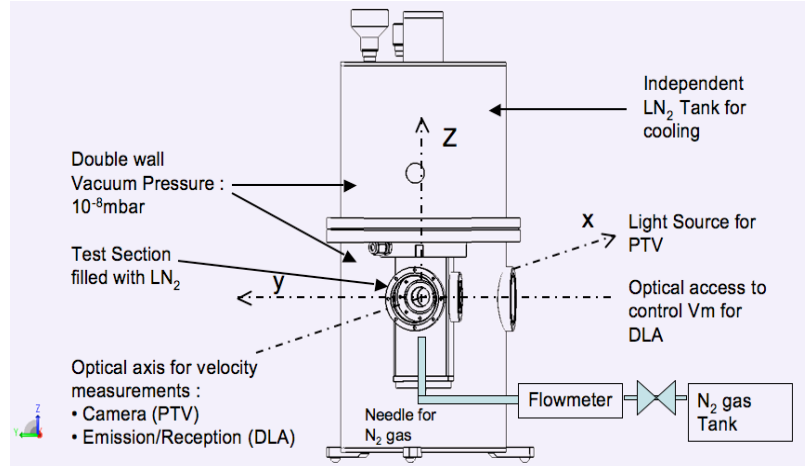


Figure 1. Sketch of the experimental setup

Diameter of a detaching bubble

N_2 gas is injected through a vertical needle positioned in the center of the test section (see Figure 1). At the outlet of the needle, the inner diameter is $D_i = 0.2mm$ and the outer diameter is $0.4mm$. The gas is injected in the test section filled with saturated liquid N_2 at T_b and P_s .

During gas injection, the growing bubble is considered as spherical. Forces acting on the growing bubble are the weight P , the buoyancy force F_a , which is 200 times greater than P and the surface tension forces F_σ maintaining the contact between the bubble and the needle. The surface tension can be expressed as $F_\sigma = \sigma \pi D_i$, with πD_i the contact length between the bubble and the needle. F_σ is in the same order of magnitude than F_a , which is coherent with $Eu \approx 1$. The drag force F_d is neglected during the growth of the bubble (quasi-static state hypothesis). When the bubble reaches a critical radius $r_{b,c}$, the bubble takes off from the needle. The balance between forces on a detaching bubble (bubble velocity $V=0$) gives the expression of $r_{b,c}$:

$$r_{b,c} = \left(\frac{3\sigma D_i}{4g(\rho_l - \rho_g)} \right)^{1/3} \quad (1)$$

g is the gravity, ρ_l and ρ_g are liquid and gas density of N_2 (see Table 1). One can find $r_{b,c} = 0.55mm$, which is in agreement with $r_{b,c}$ estimated on images of detaching bubbles (see figure 2). Thus the volume V_b of an injected bubble is $V_b = 7.10^{-10} m^3 = 0.7mm^3$.

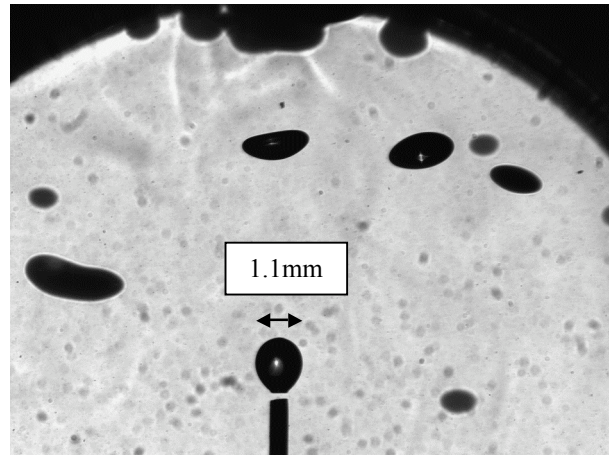


Figure 2. Image of a detaching bubble. Field of View (FOV): 10.6x14.1mm.

Upward velocity of a bubble

We consider a bubble diameter $D_b = 2r_{b,c}$, moving up with a velocity V . The velocity of the liquid is neglected against the bubble velocity. The drag forces F_d acting on the bubble are taken into account as $F_d = \rho_l C_D \pi D_b^2 V^2 / 8$. Tomiyama et al.[7] study the rising of single bubbles under a wide range of fluid properties to determine reliable correlations for drag coefficient C_D . Terminal rising velocity of single bubbles were calculated using the proposed C_D and were compared with measured data under various conditions: $10^{-2} < Eo < 10^3$, $10^{-14} < M < 10^7$ and $10^{-3} < Re < 10^{-5}$ where Eo , M and Re are Eötvös, Morton and bubble Reynolds numbers respectively. The values of various dimensionless numbers are grouped in Table 2 in the condition of the experiment presented in this paper. These dimensionless numbers are useful to consider the forces acting on the bubble. As we can see in Table 2, Re is high so the inertial forces, such as the buoyancy force, are greater than viscous forces. As $We < 8$, we can say that the bubbly flow is low velocity and can be characterized as a dripping regime [8]. Eo is similar to We but more adapted to bubbly flows. Eo is the ratio of buoyancy to surface tension forces. In our case Eo is close to unity, indicating that there is a competition between buoyancy and surface tension forces. Oh is also low indicating that surface tension forces are greater than viscous forces. Thus viscous forces have little influence on the disturbance of the bubble shape.

Tomiyama et al.[7] point out the importance of the purity of the liquid in the determination of the drag coefficient C_D . The contamination of the fluid by other particles causes modification of the motion and thus terminal velocities. In the present experiment we observe a deposit on windows and images of liquid N_2 shows contamination of the fluid by small solid particles (Estimated size $\approx 10\mu m$). Tomiyama et al. [7] also point out the influence of the shape of the bubbles in the determination of C_D . According to Clift et al.[9], bubble shapes in vertical pipes depend on the triplet (Eo , Re , M). In our case, the shape regime of bubbles is the wobbling regime, that is a non-spherical particle whose shape is close to a deformed ellipsoid. The general expression of the drag coefficient $C_{D,th}$ for a non spherical bubble in a contaminated liquid, satisfying $10^{-2} < Eo < 10^3$, $10^{-14} < M < 10^7$ and $10^{-3} < Re < 10^{-5}$ is given by equation (2):

$$C_{D,th} = \max \left\{ \min \left[\frac{24}{Re} (1 + 0.15 Re^{0.687}) \right], \frac{72}{Re} \right\}, \frac{8}{3} \frac{Eo}{Eo + 1} \quad (2)$$

Boussinesq [10] and Basset [11] studied the transient hydrodynamic forces acting on a solid sphere moving into fluid. They introduced two additional forces: the added mass force F_{AM} and the history force F_H . The added mass force is identical for any Reynolds number. The moving bubble drags a certain volume in its wake. F_{AM} results from the acceleration given to this volume when the bubble movement is non uniform and non stationary. For a spherical bubble in an infinite uniform fluid the added mass force F_{AM} is given by equation (3):

$$F_{AM} = \rho_l \frac{\pi D_b^3}{6} C_M \left(\frac{dV}{dt} - \frac{dV_l}{dt} \right) \quad (3)$$

C_M is the mass coefficient equal to 0.5 in the case of a sphere. As the liquid N_2 is quiescent in the test section, we consider that the time derivative of liquid velocity V_l is neglected.

The proper form of the history force F_H depends on the flow conditions and is currently the topic of several studies. It represents the memory of successive relative accelerations between the fluid and the particle. The history term depends on the dynamic viscosity μ_l and as $Re \approx 1000$ here (inertial forces are greater than viscous forces), we chose to neglect F_H in this model.

The second Newton law applied to the bubble leads to the following differential equation (4):

$$\frac{dV}{dt} = \left[g \left(\frac{\rho_l - \rho_g}{\rho_g} \right) - \frac{3\rho_l C_D V^2}{4D_b \rho_g} \right] / \left[1 + \frac{\rho_l}{\rho_g} C_M \right] \quad (4)$$

The analytical expression of vertical velocity V_z , derived from equation (4), is given by equation (5):

$$V_z(t) = \frac{\sqrt{a}}{\sqrt{b}} \tanh \left((t - t_0) \sqrt{a} \sqrt{b} \right) \quad (5)$$

With a , b the coefficients of the different equation under the form $dV/dt=a-bV^2$ and t_0 the time origin. Considering the drag coefficient $C_{D,th}=0.49$, obtained from equation (2), the terminal rising velocity is $V_z(t \rightarrow \infty)=0.16\text{m/s}$. However, the experimental terminal velocity is $\approx 0.19\text{m/s}$, corresponding to an effective drag coefficient $C_{D,exp}=0.35$. We can see on figure 3 the time evolution of the upward velocity of a bubble for two drag coefficients: $C_{D,th}=0.49$ with solid line and $C_{D,exp}=0.35$ with dot line.

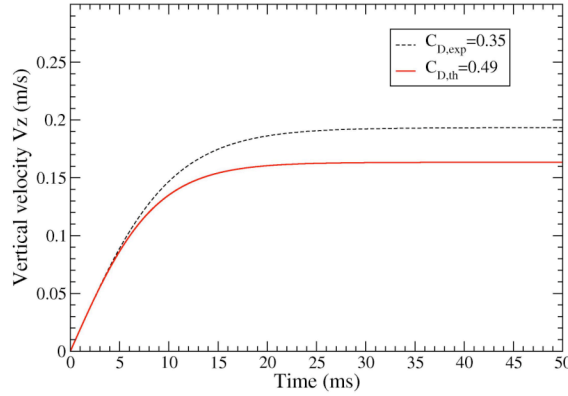


Figure 3. Analytical vertical velocity of a bubble for two drag coefficients $C_{D,th}$ (solid line) and $C_{D,exp}$ (dot line).

Influence of the medium on optical measurements

To study the feasibility of LDV in such conditions, we have to focus on the influence of the cryogenic medium on the measurement. Indeed light passes through four different medium: air, vacuum, glass and liquid N_2 whose refractive index are respectively $n_{air}=n_0=1$, $n_{glass}=1.46$ and $n_l=1.2$. We also observed that the cold medium affects some properties of light such as polarization. Moreover liquid N_2 is in saturation condition so a small amount of energy, coming from laser beams, can create nucleation bubbles. We must ensure that these phenomena do not deteriorate LDV measurements. For that purpose, we choose to compare LDV with imaging techniques, which are less sensitive to temperature gradient.

Deviation of the rays through the windows

The double wall induces the beams to pass through several windows before being scattered by the bubbles and collected by LDV. The laser beams enter the first window at room temperature with an incident angle $\alpha \approx 2.3^\circ$. This incident angle is the same when the beams reach the inner window at cold temperature. The refractive index n_l of liquid N_2 induces a deviation of the refracted ray entering liquid N_2 with an angle $\beta \approx 1.9^\circ$. The consequence is that the crossing angle of the laser beams is shifted and thus the velocity V_{meas} measured with LDV is also shifted with a multiplication coefficient 1.2 corresponding to n_l :

$$V_{cor} = \frac{V_{meas}}{1.2} \quad (6)$$

Where V_{meas} is the measured velocity and V_{cor} is the corrected velocity of a bubble in liquid N_2 . V_{meas} is thus overestimated in comparison with V_{real} .

We checked that the crossing of laser beams was not modified when light goes through the entire cryostat. That is the crossing of the 4 laser beams is preserved before entering the cryostat, at room temperature, as well as after leaving the cryostat, when the test section is filled with cold N_2 gas at $T=86\text{K}$.

Influence on polarization of light

The measurement volume of LDV is constituted of interference fringes, created by the crossing of two lasers of same wavelength λ and same polarization. A high contrast between bright and dark fringes ensures a good signal to noise ratio to analyze the signal scattered by particles passing through LDV measurement volume. According to Albrecht et al.[12] a loss of polarization of light induces a decrease of the contrast of the fringes.

We noticed a modification of the light polarization when the laser beam passes through the cryostat. Indeed we checked that the polarization of light emitted by the LDV system was polarized horizontally and parallel to

the diffusion plane. If the whole cryostat is kept at room temperature (filled with ambient air), the light polarization remains unchanged when laser beams cross over the cryostat, which is through ambient air and multiple window accesses. However when the 2L tank is filled with liquid N₂ at T_b , and the test section is filled with N₂ gas at $T=86K$, we observe a change for some beams and almost no polarization for others. Unfortunately, we did not find a particular orientation of the light polarization living the cryostat at cold temperature, even for different incident polarization. We cannot state on the reason of this change of light property but we think that the polarization is modified either by the windows and/or by small contamination particles seeding. Indeed windows are facing important temperature gradients inducing high mechanical constraints and yielding to an anisotropic medium. Moreover convection movements can animate the contamination particles, observed in liquid N₂ at T_b , when the test section is filled with N₂ gas at 86K and cause modification of light polarization.

Optical setup of velocimetry techniques

Laser Doppler Velocimetry (LDV)

A commercial Dantec LDV system has been used for 2D velocity measurements. An Argon laser delivers green ($\lambda_g = 514.5$ nm) and blue ($\lambda_b = 488$ nm) beams. The beam separation is 40 mm and the focal length of the transmitter lens is $f = 500$ mm. The polarization of the laser beams is parallel to the diffusion plane.

The light scattered by bubbles is collected in backward scattering configuration. This particular scattering mode was chosen to be close to the experimental setup on a turbo pump, where optical access is limited. Moreover LDV measurements are set to coincidence mode. Thus the velocity measurement of a bubble is validated only if the two components of the velocity provide a valid measurement. It means that signals analyzed by the processor unit must be validated on the two velocity channels. Therefore the data rate of the coincidence setup is generally smaller than without coincidence.

The measurement volume of LDV is an ellipsoid of revolution of diameter $150\mu m$ and $3.75mm$ in length. The diffraction pattern is constituted of 23 fringes separated of $6.436\mu m$.

For a given position of the measurement volume, the total number of validated bubbles is set to 500 when the data rate is low ($\approx 1Hz$ for boiling bubbles) and is set to 2000 bubbles when the data rate is relatively high ($\approx 10Hz$ for injected bubbles).

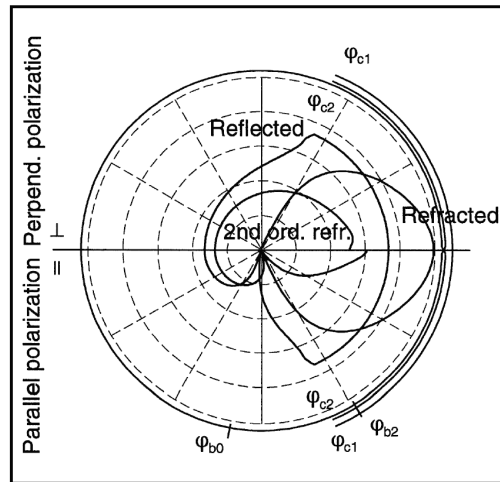


Figure 4. Scattering diagram of air bubble in freon.
Angular dependence on intensity (logarithmic scale, one decade per dotted circle)
for perpendicular and parallel polarization. $n_{rel}=0.83$ [12].

The choice of the collection angle is an important parameter to analyze the scattered light. Indeed, the contribution of the different scattering modes such as reflection, refraction and 2nd order refraction can superpose and give a spurious interpretation of the scattered light. The scattering diagram presented on figure 4 provides the angular distribution of the relative intensity of the scattering light in each of the three scattering modes. The calculations are based on geometrical optics and were made for an air bubble of diameter $50\mu m$ in freon and a single point receiving aperture [12]. This case is close to the one studied here as the relative index of the medium $n_{rel}=n_g/n_l=0.83$ in both cases. As we can see on figure 4, the best collection angle for the reflection mode is $\varphi_{b0}=100.4^\circ$. The collection angle in the present paper is set to 180° thus reflection and 2nd order refraction are relatively close in intensity. Blondel [6] showed that LDV measurements in reflection mode are feasible in a medium of relative refractive index $0.8 < n_{rel} < 1$.

We noticed that laser power has to be low ($\approx 15\mu\text{W}$ for each laser beam) to avoid the saturation of the photomultipliers. Saturation is probably due to multiple internal reflections inside the cryostat due to the numerous windows. Moreover, internal walls and the needle are metallic so they scatter a large amount of light that can also induce the saturation of the photomultipliers.

High-speed Imaging

The high-speed imaging system is constituted of a monochrome camera Phantom V12.1 from Vision Research and an incoherent and continuous light source. A Nikon objective of focal length 180mm and an extension tube of 52.5mm are mounted on the camera. The camera is equipped with a 8-bit CCD sensor of maximum resolution 1280 x 800 square pixels of side $20\mu\text{m}$. The maximum frame rate at maximum resolution is 6 000 frames/s. For the present study, the frame rate is set at 5 000 frames/s, the exposure time is $90\mu\text{s}$ and the resolution is 146p/cm for 510 x 381 pixels.

Single bubbles in movie files are chosen manually: bubbles are labelled with a single point, which is tracked through each image until it disappears from the image field. Each bubble rising is tracked automatically with the TEMA Motion software from Photsonics. Thus the temporal evolution of the position and the velocity of the bubble can be obtained.

Particle sizing and Tracking Velocimetry (PTV)

The PTV system is composed of a monochrome camera JAI CV M2 equipped with a 10-bit Kodak CCD sensor (KAI-2020M) composed of 1200x1600 square pixels of side $7.4\mu\text{m}$. A telecentric objective (S5LPJ1005 Sill Optics Correctal) is mounted on the camera. The specificity of this objective is a constant magnification (0.84) over a wide range of focus. The field of view (FOV) is 10.6mm x 14.1mm with a resolution of 113.2 pixels/mm. The working distance is 123mm and the f-number has been set to 5.6. The camera is set to PIV mode to record pairs of images with very short interval. The minimum time interval between two images is $1.5\mu\text{s}$.

The bubbles are illuminated in a backlight configuration by a non-coherent short flash source (Nanotwin-HSPS, duration 12ns) to freeze the bubbles on the images. This light source delivers a double spark synchronized with the shutter of the camera.

The PTV system is based on a drop-size measurement technique by imaging [13]. The drop-sizing algorithm first detects the bubbles in the images, then each image bubble is analyzed by image processing. The bubble size, shape and coordinates of the gravity centre of the projected area of the image bubble are provided with these programs.

500 couples of images are recorded to obtain a statistically robust drop-size distribution. We show on figure 5 the probability density function (*pdf*) of the bubbles in the cryostat, that is for boiling bubbles and injected bubbles. We can see that the drop-size is almost monodisperse around a bubble diameter of 0.9mm ($D_{20}=0.996\text{mm}$).

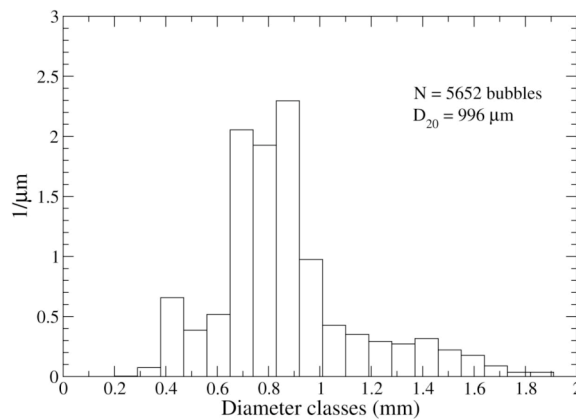
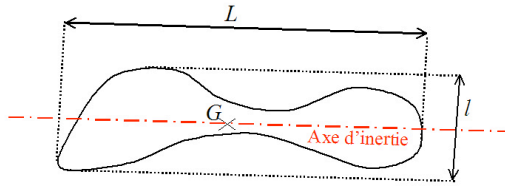
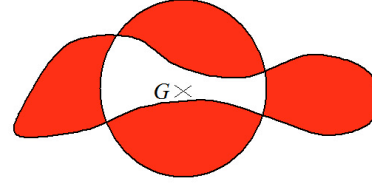


Figure 5. *pdf* of the bubbles in the test section.

As bubbles are large and non-spherical we define 3 parameters to characterize bubbles:

- The mean diameter D_{20} , characterizing the size of a bubble [14]. D_{20} is the diameter of the bubble whose area is the mean area of all the bubbles.
- The Ellipticity ε characterizes the shape of a bubble [15]. The Ellipticity ε is the ratio of the maximum length L_{max} along the inertial axis of the particle image to the minimum length L_{min} perpendicularly to this inertial axis (See figure 6.A and equation 7). This parameter ranges between 0 and 1. It is equal to 1 for a disc.

$$\varepsilon = \frac{L_{\min}}{L_{\max}} \quad (7)$$

A) Definition of the Ellipticity ε .B) Definition of the Sphericity Sp **Figure 6.** Definition of the morphological parameters

- The Sphericity Sp , also characterizes the shape of a bubble [15]. Sp is based on the equivalent disc, i.e. the disc having the same area than the object and centered on the gravity centre of the object. The Sphericity Sp is defined by equation 8. It corresponds to the ratio of the difference between the union and the intersection of the surface S of the particle and the surface S_C of the equivalent disc and the surface S of the particle. It can be seen as the ratio of the coloured surface on the Figure 6.B to the surface of the particle S . This ratio is null in the case of a sphere and limited to 2 for elongated shapes.

$$Sp = \frac{A(S \cup S_C) - A(S \cap S_C)}{A(S)} \quad (8)$$

In equation 8 $A(x)$ stands for the area of surface x . High values of Sp refers to chains of droplets or ligaments in general.

The mean morphological parameters for bubbles in the test section are $Sp=0.4$ and $\varepsilon=0.57$. The large value of Sp and the low one for ε confirm that these bubbles are not spherical. The mean number of droplet is 5.6 bubble/image of mean diameter $D_{20} \approx 1\text{mm}$.

The PTV system, providing the velocity of bubbles, is based on the Malek's algorithm [16], developed for characterizing 3D flows recorded by holography. This algorithm has been adapted to 2D flows in our case. 2D coordinates of bubbles from a double exposure image are the input of the algorithm. The algorithm finds the best matching between the two sets of points. The output of the algorithm is the displacement of each matched particle in the image. Individual bubble velocities are obtained knowing the time interval Δt between two images.

The time interval Δt is a relevant parameter to ensure the best possible matching. Fdida [17] suggests a criterion to choose the optimum Δt for a given flow. Considering Tz as the distance covered by the particle between a couple of images and Δd the mean distance between two particles in a single image, the following criterion must be verified: $0.4 < Tz/\Delta d < 0.8$ [17]. In our case, $\Delta d \approx 1\text{mm}$, which represents the mean distance between two rising bubbles in images. Thus one can find $0.4\text{mm} < Tz < 0.8\text{mm}$. The terminal velocity is $\approx 0.2\text{m/s}$, so the optimum time interval Δt is: $2\text{ms} < \Delta t < 4\text{ms}$. Thus we choose $\Delta t=3\text{ms}$.

Grouping together all the displacement vectors obtained with 500 couples of images, it leads to a vector field filling the entire image (see figure 7). A Region of Interest ROI is then defined to extract only the displacement of the bubbles created by gas injection. A dot line on figure 7 illustrates the ROI. Only bubbles whose gravity centre is located into the ROI are retained. The mean number of droplets in ROI is $\approx 3\text{bubbles/image couple}$.

Outside from the ROI, the vector field show a global ascending movement (see figure 7). Thus no particular convection movement is observed in the test section. Inside the ROI, the rising velocity of the injected bubbles can be obtained. However we can see on some images, bubbles coming from the natural boiling, and entering the ROI and can be taken into account in the data.

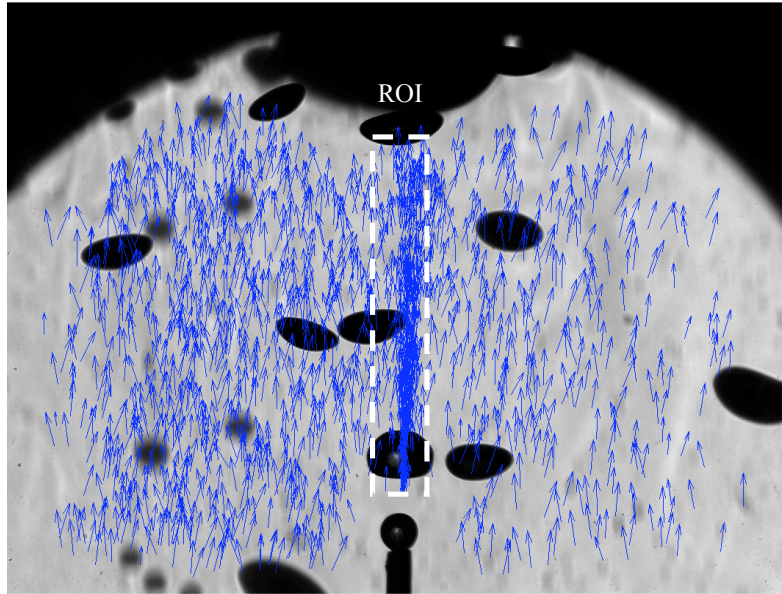


Figure 7. PTV vector field in the test section obtained by 500 couples of images. (FOV 10.6 x 14.5mm)

Results and discussion on velocity measurements

Bubble generation with gas injection

A Kobold flowmeter model KFR with a suspended float regulates the flow rate. The flow meter is not enough accurate to control precisely the flow in a “dripping” regime. The range of the flow meter is $1.7 \cdot 10^{-7} < Q < 8.3 \cdot 10^{-6} \text{ m}^3/\text{s}$ with a precision of $3.3 \cdot 10^{-7} \text{ m}^3/\text{s}$. Velocity measurements are made on a stable “dripping” regime of the flow, with the flow meter set at its minimum. As we could not measure the flow rate with the flow meter, we estimated the effective flow rate with high-speed imaging, by measuring the generation frequency of bubbles ($\approx 24\text{Hz}$). This frequency corresponds to a stable flow rate $Q \approx 1.7 \cdot 10^{-8} \text{ m}^3/\text{s}$.

Study of boiling bubbles

In the test section, liquid N_2 is at boiling state at atmospheric pressure and T_b . Most of boiling bubbles are created near the internal windows and at the bottom of the test section. We notice that boiling bubbles created at the bottom of the test section have a size, a shape and a behavior close to the bubbles injected by gas. On the other hand, bubbles created near the window are smaller ($D_b \approx 300 \mu\text{m}$), almost spherical and we observe that they grow in size during their rise. Their terminal velocity is smaller $V_\infty \approx 0.1 \text{ m/s}$ whereas for large boiling bubbles $V_\infty \approx 0.21 \text{ m/s}$. Thanks to the small measurement volume of LDV, it can be located above the needle and LDV measures only the velocities of the bubbles generated by N_2 gas injection at the centre of the test section.

Location of measurement points

We study the rising of a bubble in the (O,y,z) frame where the origin O is the outlet of the needle. Z axis is vertical and Y axis is horizontal as illustrated on figure 8. Acquisitions with the three velocimeters are not recorded simultaneously but successively in the stable flow regime mentioned above. In order to compare the results of the three velocimeters, measurement points are located as follows:

- For PTV and high-speed imaging, only bubbles in the ROI are considered ($1.1 \text{ mm} < Z < 7 \text{ mm}$). The lower limit of the ROI corresponds to the formation zone of the bubble. The upper limit is due to the presence of a N_2 gas “pocket” trapped in the flange of the window port.
- LDV measurement points have been defined inside the interval $0.6 \text{ mm} < Z < 7 \text{ mm}$. For measurement points below $Z = 0.6 \text{ mm}$, the needle reflects too much light so the photomultipliers of the LDV system are saturated. For $0.6 \text{ mm} < Z < 2 \text{ mm}$, measurement points are regularly spaced of 0.1 mm and for $2 \text{ mm} < Z < 7 \text{ mm}$, the spacing interval is 0.25 mm . To ensure the location of the measurement volume of LDV, images are made through the third optical access of the cryostat.

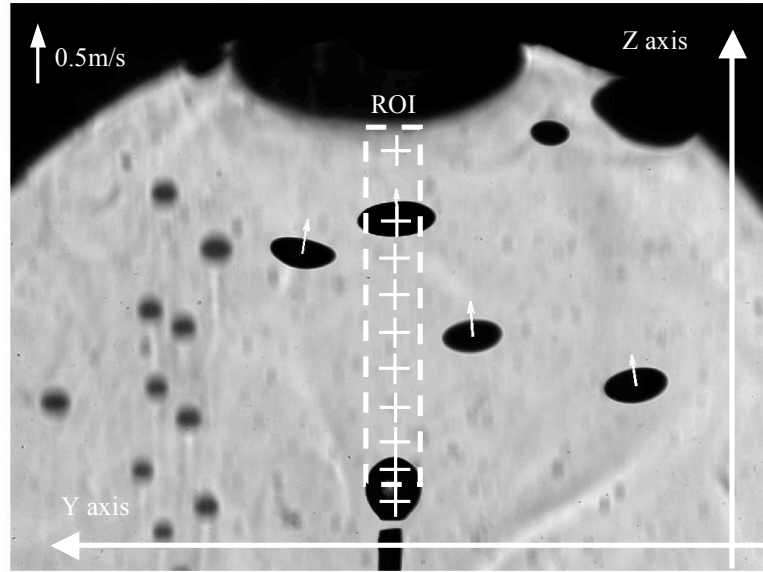


Figure 8. Location of measurement points for LDV (cross symbols), PTV and high-speed imaging (ROI in dot line). FOV :10.6x14.1mm.

Comparison between velocity measurements

High-speed Imaging measurements

We present on figure 9 velocity measurements obtained with high-speed imaging. The bubble frequency is stable at 24Hz which corresponds to a flow rate $Q=1.7 \cdot 10^{-8} \text{ m}^3/\text{s}$. The temporal variation of the vertical velocity for 5 successive bubbles is presented on figure 9. First we can see sinusoidal fluctuations of the velocity which were not expected by the model (in dot lines on Figure 9). The frequency of these oscillations is $\approx 125\text{Hz}$ and has been observed on all bubbles tracked on high-speed imaging movies.

These oscillations are probably due to the fluctuations of the shape of the bubbles. High-speed imaging movies show that when the bubble detaches from the needle, its shape is flat at the bottom of the bubble (see figure 9). This causes an oscillation of the shape of the bubble during its rising. As the drag coefficient depends on the shape of the bubble, the upward velocity is thus affected by the oscillation of the shape of the bubble.

The bubble's fundamental frequencies can be obtained from the classic analysis by Rayleigh-Lamb. The natural frequency of a droplet, that is for the mode $m=2$, is defined by equation 9. We found that the natural frequency for a bubble $D_b=1.1\text{mm}$ is $f_2=140\text{Hz}$, which is in agreement with the measured oscillation frequency of bubble velocity. This mode is clearly illustrated on images at the top of figure 9.

$$f_m = \sqrt{\frac{2\sigma(m+1)(m-1)(m+2)}{\pi D_b^3 (\rho_g(m+1) + \rho_l m)}} \quad (9)$$

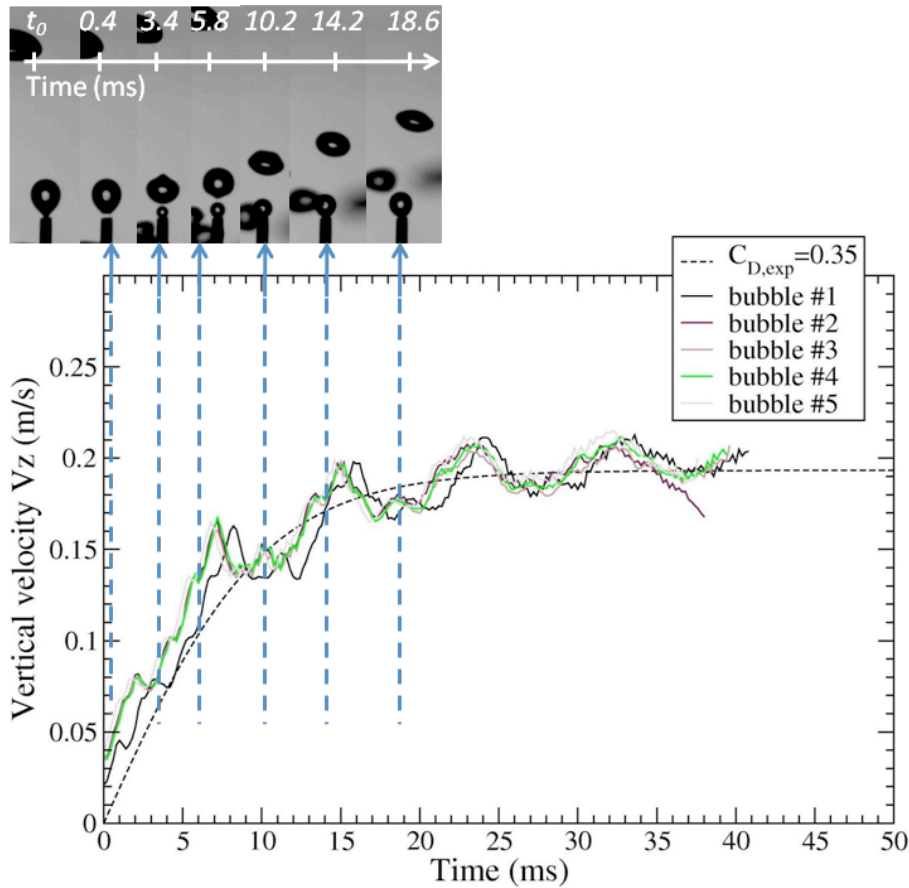
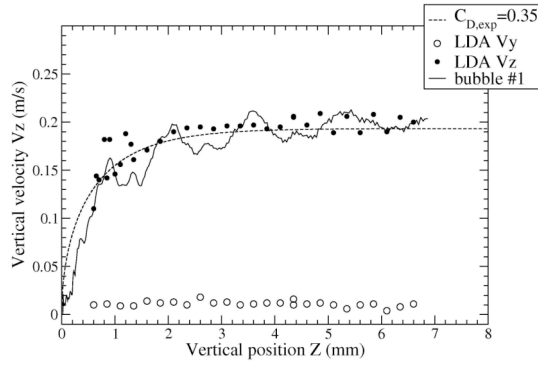


Figure 9. Velocity measurements of five successive bubbles (solid lines).
Comparison with the model (dot line)

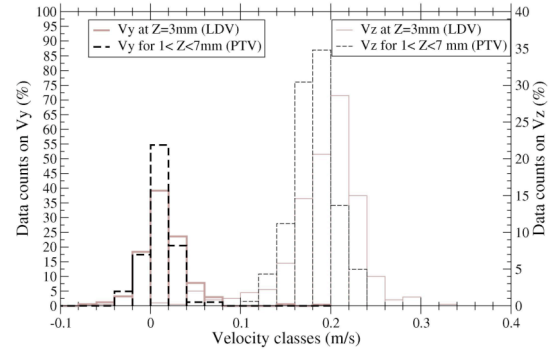
LDV Measurements

Figure 10.A shows a good agreement between vertical velocities measured by LDV, high-speed imaging and the model. We saw in section “deviation of the rays through the windows” that LDV measured velocities must be corrected but results shown on Figure 10 are not corrected. It is probably due to the presence of boiling bubbles in the measurement volume of LDV. Indeed, we notice two populations of bubbles of different velocities for the positions $Z < 1.1\text{ mm}$. We observe a population of bubbles whose vertical velocity $\approx 0.05\text{ m/s}$ corresponding to the injected bubbles, and a population of bubbles whose rising velocity is $\approx 0.22\text{ m/s}$ corresponding to the terminal velocity of boiling bubbles coming from the bottom of the test section. Indeed, for this position close to the needle, injected bubbles may not have such a high velocity. As boiling bubbles and injected bubbles have similar size, it is impossible to clearly distinguish the two populations of bubbles as soon as $Z > 1.1\text{ mm}$. We can conclude that LDV measurements are biased because of the presence of two populations of bubbles in the measurement volume of LDV: boiling and injected bubbles. The presence of boiling bubbles in the test section is reinforced by LDV measurements without gas injection. In this case, velocity measurements show a low data rate ($\approx 1\text{ Hz}$) and a single peak on velocity histograms centred on $V_z = 0.2\text{ m/s}$, even close to the needle outlet ($Z < 2\text{ mm}$).

We show on Figure 10.B velocity histograms obtained with LDV for the measurement point $Z = 3\text{ mm}$ (solid lines). Horizontal velocities are nearly centred on $V_y = 0\text{ m/s}$, thus we can assume that bubble trajectory is vertical. The vertical velocities are centred on $V_z = 0.2\text{ m/s}$ which is in agreement with the model.



A) Velocity measurements with LDV (empty symbols for V_y and filled symbols for V_z), high-speed imaging (solid line) and the model (dot line).



B) Comparison of velocity histograms obtained with PTV (dark dot line for V_y and smooth dot line V_z) and LDV (dark solid line for V_y and smooth solid line V_z).

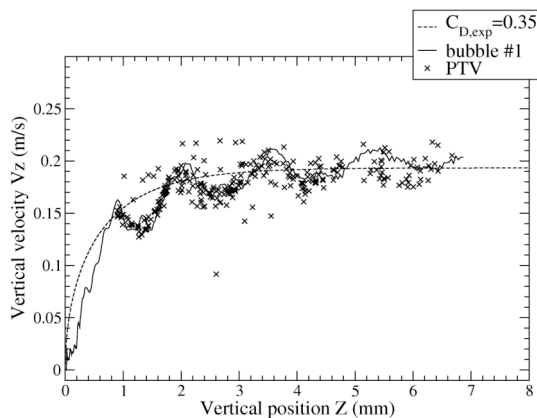
Figure 10. Analysis of LDV measurements

PTV Measurements

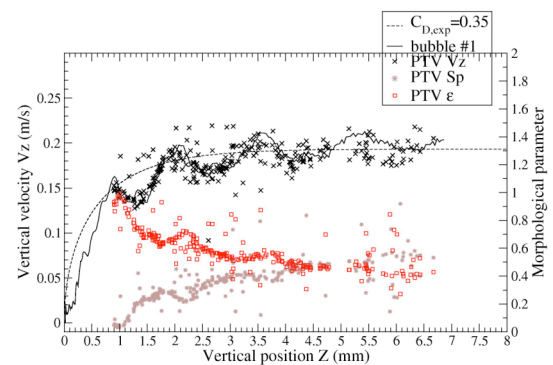
On figure 11, PTV measurements (cross symbols) are compared with vertical velocities obtained for a single bubble with high-speed imaging (solid line) and the model (dot line). We can say that PTV measurements are in a good agreement with high-speed imaging. Measurements shown on Figure 11 are obtained from a bubble sample of 273 bubbles, located in the ROI. As for LDV, we notice spurious points with a velocity ≈ 0.2 m/s and also spurious points with an abnormal low velocity probably due wrong vectors resulting from the point-matching algorithm. On figure 10.B, we can see that PTV velocity histograms (dot lines) are in agreement with the one of LDV (solid line) for both V_y and V_z . Differences are probably due to the measurement volumes of the two techniques. Indeed, the bubbles sample of PTV group together data inside ROI whose size is larger than the measurement volume of LDV.

Moreover LDV and PTV face similar difficulties to detect the oscillations of the vertical velocity far from the needle, that is for $Z > 4$ mm. This can be due to the fact that measurements are not synchronized with the bubble detaching from the needle, which is hard to detect experimentally. Thus far from the needle, bubble velocity fluctuations are not totally in phase.

We saw in the section “High-speed Imaging measurements” that bubble shape seems to be responsible of velocity fluctuations. We present in figure 11.B the evolution of the velocity and the shape parameters Sp and ε (respectively star and square symbols). We can see that bubble shape is nearly spherical when detaching ($Sp \approx 0$ and $\varepsilon \approx 1$). During the rising, the bubble shape moves away from the spherical shape and tends towards an ellipsoidal shape $Sp \approx 0.5$ and $\varepsilon \approx 0.5$. This is in agreement with the predicted bubble shape [9]. Moreover, morphological parameters Sp , ε and the vertical velocity show similar fluctuations particularly for $Z < 4$ mm. This reinforces the hypothesis that fluctuations of bubble velocity are due to oscillations of the bubble shape.



A) Velocity measurements with PTV (cross symbols), high-speed imaging (solid line) and the model (dot line).



B) Velocity and shape measurements with PTV (symbols), high-speed imaging (solid line) and the model (dot line).

Figure 11. Analysis of PTV measurements

Conclusion

We show in this paper the feasibility of LDV measurements in liquid N₂ at T_b . Even with a low signal to noise ratio, due to loss of polarization, comparisons with other velocity measurements by imaging show a good agreement between 3 techniques. Bubble shape characterization, obtained with imaging, represents valuable information to explain the oscillations of velocity.

Differences are due to the measurement volume sizes of each technique and also due to a lack of control of the flow (flow rate and presence of boiling bubbles). Indeed, seeding bubbles are not adapted to LDV measurement as their size is large and their shape is not spherical. Moreover, the influence of the laser beam in the medium must be evaluated, that is to check if bubbles can be created within the LDV measurement volume and understand the change of polarization due to the crossing of the cryostat.

The perspective of this work is to modify the seeding of bubbles. A heated wire of diameter 100µm will be immersed in liquid N₂. With a low electric power and a sinusoidal control of the intensity in the wire, a nucleation regime can be obtained. This kind of regime would be more adapted to LDV measurements.

Acknowledgement

This work is performed in the frame of the LAPLACE project, led by Snecma (France), with financial support of Région Haute-Normandie and European Union (ERDF).

Table 1. Properties of Nitrogen Liquid and Gas

Symbol	Signification	Value	Units
T_b	Boiling temperature at P_s	77.3	(°K)
σ	Surface tension at T_b	$8.8 \cdot 10^{-3}$	(N/m)
ρ_L	Density of liquid at P_s	810	(kg/m ³)
ρ_G	Density mass of gas at T_b and P_s	4.59	(kg/m ³)
μ_L	Dynamic viscosity of liquid at T_b	160	(µPa.s)
μ_G	Dynamic viscosity of gas at T_b	5	(µPa.s)
n_L	Refraction index of liquid at T_b and $\lambda=589\mu\text{m}$	1.2	Dimensionless
n_G	Refraction index of gas at $T=273^\circ\text{K}$	1.0	Dimensionless
P_s	Vapour saturation pressure at T_b	101.3	(kPa)

Table 2. Several dimensionless numbers for the experiment.

Symbol	Name	Expression	Signification	Value
Eo	Eötvös	$Eo = \frac{g(\rho_l - \rho_g)D_b^2}{\sigma}$	Buoyancy force/surface tension force	0.89
We	Weber	$We = \frac{\rho_l V^2 D_i}{\sigma}$	Inertial force/surface tension force	0.74
Re	Reynolds	$Re = \frac{\rho_l V D_b}{\mu_l}$	Inertial force/viscous force	1000
M	Morton	$Eo = \frac{g\mu_l^4(\rho_l - \rho_g)}{\rho_l^2 \sigma^3}$	Property group of the two phases	$1.1 \cdot 10^{-11}$
Oh	Ohnesorge	$Oh = \frac{\mu_l}{\sqrt{\rho_l \sigma D_b}}$	Viscous force/surface tension forces	$1.9 \cdot 10^{-3}$

Nomenclature

a	Coefficient [$\text{m}\cdot\text{s}^{-2}$]
$A(x)$	Area of surface x [m^2]
b	Coefficient [s^{-1}]
C_D	Drag coefficient [dimensionless]
C_M	Mass coefficient of F_{AM} [dimensionless]
D_b	Bubble diameter [mm]
D_i	Inner diameter of the needle [mm]
D_{20}	Mean diameter [μm]
f	Focal length [mm]
f_m	Oscillation frequency of a bubble [Hz]
F_a	Buoyancy force [N]
F_{AM}	Added mass force [N]
F_d	Drag force [N]
F_H	History force [N]
F_σ	Surface tension force [N]
g	Gravity [$\text{m}\cdot\text{s}^{-2}$]
i	Fringe spacing [mm]
L_{min}	Maximum length along the inertial axis of a bubble
L_{max}	Minimum length along the inertial axis of a bubble
m	Mass [kg]
n_{air}	Refractive index for air [dimensionless]
n_0	Refractive index for vacuum [dimensionless]
n_g	Refractive index for N_2 gas [dimensionless]
n_{glass}	Refractive index for glass [dimensionless]
n_l	Refractive index for liquid N_2 [dimensionless]
n_{rel}	Relative refractive index [dimensionless]
P	Weight [N]
P_s	Vapor pressure of N_2 [kPa]
Q	Volumetric flow rate [m^3/s]
$r_{b,c}$	Critical radius of a bubble [mm]
ρ	Density [$\text{kg}\cdot\text{m}^{-3}$]
S	Surface of a bubble [m^2]
S_c	Surface of the equivalent disc [m^2]
Sp	Sphericity of a bubble [dimensionless]
t	Time [s]
Tz	Distance covered by a particle [m]
V_b	Volume of a bubble [m^3]
V	Bubble velocity [m/s]
V_{meas}	Velocity measured by LDV [m/s]
V_{real}	Corrected velocity [m/s]
α	Incident angle [rad]
β	Incident angle [rad]
Δd	Mean distance between droplets [m]
Δt	Time interval between double exposure images [s]
ε	Ellipticity of a bubble [dimensionless]
λ	Wavelength [nm]
μ	Dynamic viscosity [Pa.s]
σ	Surface tension [N/m]

Subscripts

g	gas
l	liquid
x,y,z	refers to x,y,z axis

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